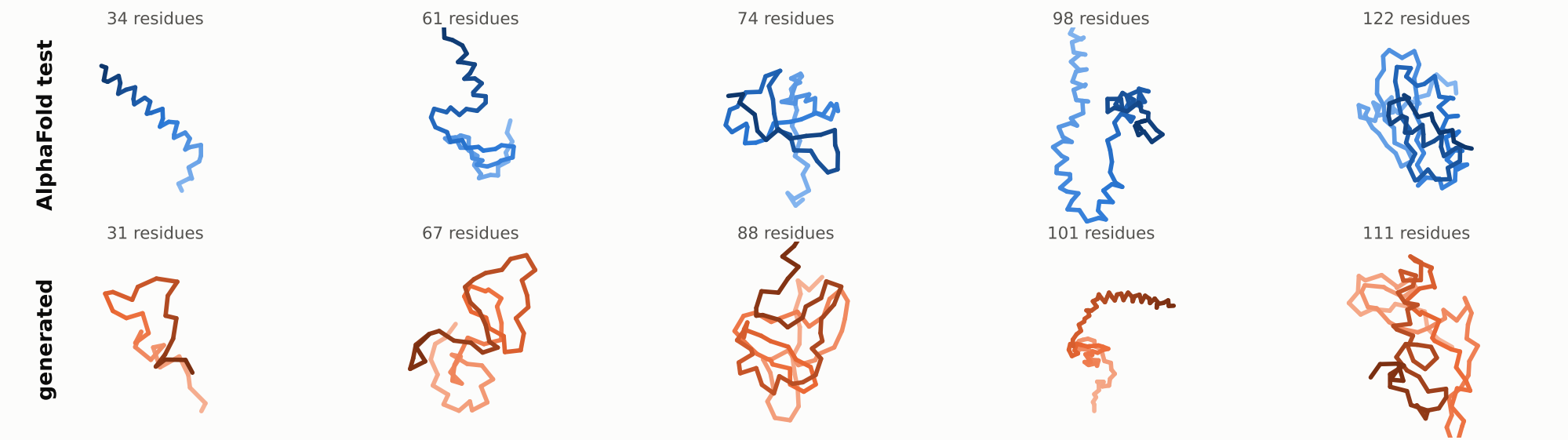
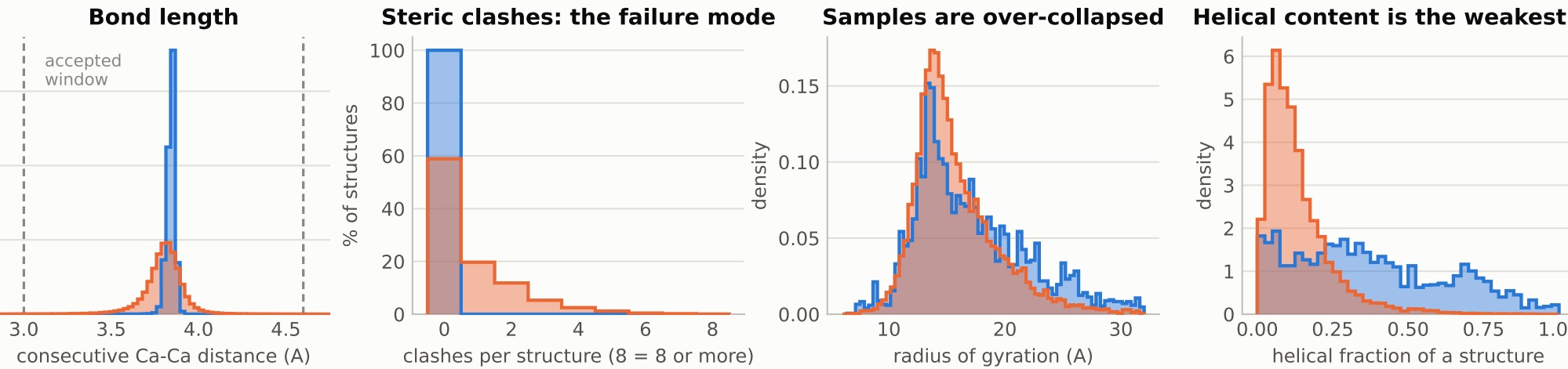
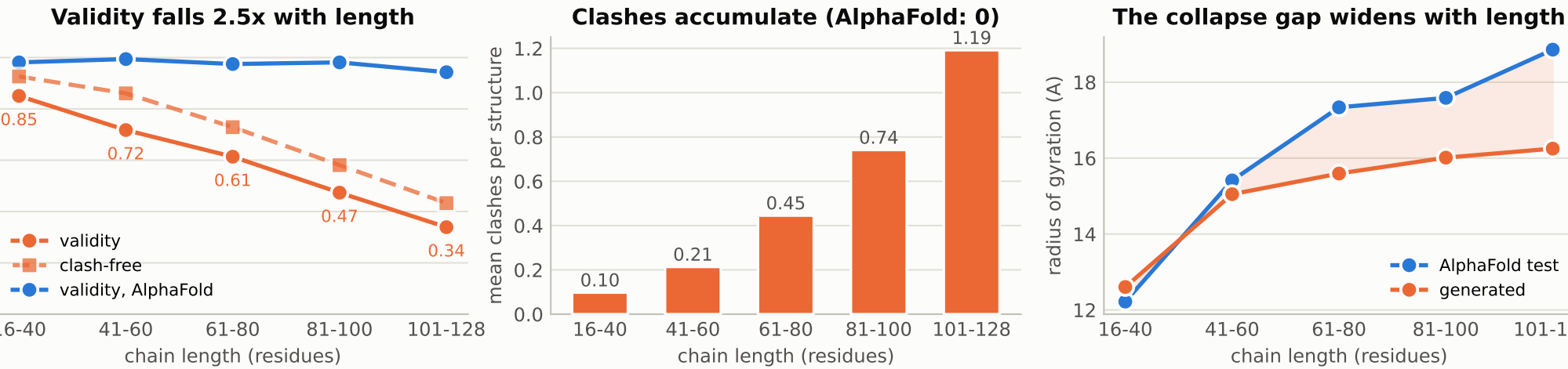
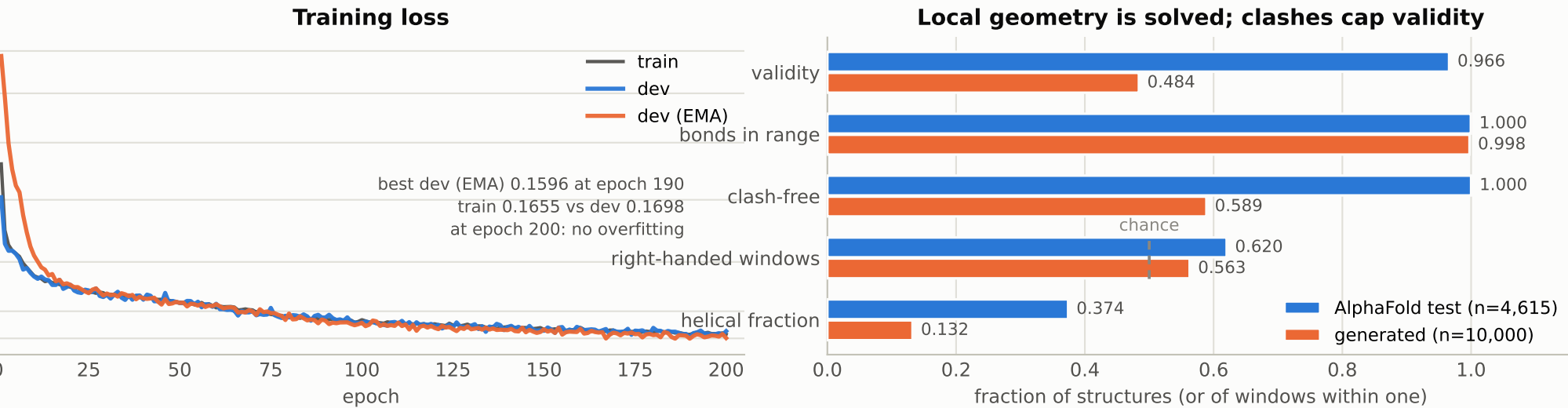


Unconditional Ca-backbone diffusion on SwissProt-Pfam

4.16M-parameter sparse SE(3) EGNN, 9 layers, hidden 256, k=16 neighbours | 37,476 train / 4,728 dev / 4,615 test chains of 16-128 residues, single Pfam family, AlphaFold DB structures
200 epochs at batch 128 in 2.13 h on one A100-40GB | 10,000 backbones sampled with 1000 denoising steps | every metric computed identically for both rows, so the AlphaFold row is the ceiling



Ca traces above: clash-free examples matched on length, shade running N to C terminus. Bonds are solved and handedness is learned; clashes and missing helices are what is left.